Combining agile and plan-driven methodologies for managing complex IT projects: towards three hybrid models

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Abstract: Although previous research works have shown interest in combining agile methodologies with plan-driven project management, no defined hybrid models have been suggested for managing complex projects. Teams lack structured processes that help them integrate agile principles and practices in their plan-driven environment. Given this focus, our paper aims at filling this gap by proposing hybrid models that assists teams in managing complex projects. A qualitative research approach has been adopted. Eighteen semi-structured interviews were conducted. Data analysis highlights three major hybrid models. For each hybrid model, the associated benefits, challenges and success factors were analysed and explained.

Keywords: agile methodologies; plan-driven methodologies; hybrid approach analysis; challenges; success factors.

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1 Introduction

Over the past few decades, several different methodologies for guiding teams in successfully managing IT projects have emerged. These methodologies differ in terms of project management principles, processes, practices and tools. While some rely on linear lifecycles with extensive documentation and up-front planning, others embrace short-delivery lifecycles with iterative planning and continuous adaptation to changes. The former is widely viewed, by the academic literature, as more traditional, structured methodologies, whereas the latter is seen as 'lightweight' methodologies with a strong emphasis on constant collaboration and feedback between team members and client. These lightweight or 'agile' methodologies have gained a lot of attention among business and IT departments working in dynamic and competitive environments (Mckinsey, 2021; Bianchi et al., 2020; Iskanius et al., 2006). They are perceived as a better way to increase in customer satisfaction, driving innovation and cope with evolving technologies and changing customer demands (Mckinsey, 2021; Digital.ai, 2021; Findsrud, 2020; Rauch et al., 2017; Petersen and Wohlin, 2009; Paasivaara et al., 2008).

An increasing number of surveys and research papers have been conducted on both traditional and agile project management methodologies (Mckinsey, 2021; Digital.ai, 2021; HP Enterprise, 2017). Research studies have highlighted the characteristics of these two project management methodologies as well as their impact on project teams (Sutherland et al., 2007; Khalil et al., 2013; Cooper et al., 2002). They also focus on the success factors and failure risks associated with each. Traditional methodologies such as waterfall and 'V' cycle models appear to be more appropriate for critical projects that necessitate extensive documentation and up- front detailed planning (Fernandez and Fernandez, 2008; Coram and Bohner, 2005). Meanwhile, agile methodologies seem to be more suited for innovative projects where requirements are unpredictable and unstable (Poppendieck and Poppendieck, 2006). Such projects require constant adaptation to changes, as well as knowledge sharing and knowledge creation between team members (Khalil and Khalil, 2016; Khalil et al., 2013; Lloyd-Walker and Walker, 2011; Paasivaara et al., 2008; Sutherland et al., 2007).

Even though innovative projects value iterative development, constant feedback and adaptation to changing requirements exist, successful management of such projects can necessitate more structured approaches based on formal documentation, initial project definition and planning (Battistella et al., 2017; Bartsch et al., 2013). This is even more important when projects are complex, involve different teams and stakeholders, and are characterised by high security and technical levels (Nemkova, 2017; Leimeister, 2010). In this respect, a hybrid approach that combines agile and traditional project management methodologies can be useful for teams involved in such projects. However, how can agile and traditional methodologies be mixed? Which challenges and success factors should be considered while blending these project management methodologies?

Although several studies have shown interest in combining agile methodologies with plan-driven project management approaches, a few have suggested using hybrid models when managing complex projects (Rodov and Teixidó, 2016; Sommer al., 2015; Lozo and Jovanovic, 2012). Besides, the suggested hybrid models found in the literature are still insufficiently known within organisations, necessitating further investigations.

Given the lack of research findings on hybrid project management methodologies, this paper aims to fill this gap by proposing a hybrid model that can be deployed by organisations and project teams. The success factors, challenges and benefits associated with the proposed hybrid model will be highlighted and analysed. The suggested model results from the analysis of a set of semi-structured interviews conducted with eighteen consultants who specialise in both project management and change management.

We structured our paper as follows: Section 2 compares agile and traditional project management characteristics and reviews the related work regarding their combination. Section 3 describes the research methodology which we used to identify and describe the proposed hybrid models. Section 4 presents the hybrid model and associated benefits, risks and success factors. We conclude this work with a discussion on the major implications and limitations of this work.

2 Agile and plan-driven methodologies: related works

2.1 Agile vs. plan-driven methodologies

While many different project management methodologies exist, the two main types are traditional and agile methodologies. This section portrays the major principles on which agile and traditional methodologies rely and review the major findings of related works.

Traditional projects proceed through clearly defined phases, with the product deliverable being produced at the end of the process (Bianchi et al., 2020; Cicmil et al., 2006; Loos et al., 2011). Extensive planning and documentation are valued throughout the whole process. The project's requirements are defined up-front in order to minimise uncertainty and therefore better manage risks (Lange, 2021; Cooper and Sommer, 2016). In this respect, the development team and the customer agree on what will be delivered early in the development process. As the scope of the work is defined in advance, project progress is easily monitored (Cooper and Sommer, 2016). For Fernandez and Fernandez (2008) and Coram and Bohner (2005), traditional approaches seem more appropriate for stable projects with clear initial user requirements and goals. Up-front planning gives support for people who are sponsoring the project (Karlström and Runeson, 2005). However, many practitioners have criticised traditional project management methodologies for being rigid and unable to adapt to inevitable changing demands

(Highsmith and Cockburn, 2001; Poppendieck and Poppendieck, 2006; Petersen and Wohlin, 2009). In addition, excessively specific and detailed plans and extensive documentation are viewed as wasteful as they become obsolete and must be reworked (Bianchi et al., 2020; Petersen et al., 2009; Poppendieck and Poppendieck, 2003, 2006; Cicmil et al., 2006; Boehm, 2002).

Meanwhile, agile methodologies emphasise short and iterative development lifecycles, ongoing collaboration between the development team and the client, and adaptation to changing requirements (Nemkova, 2017; Vázquez-Bustelo and Avella, 2006). These values and principles are established with a set of ceremonies (retrospective meetings, daily meetings, weekly meetings with the client), practices (iterative development, pair programming, collective code-ownership, continuous code integration, unit-testing…), and tools (product backlogs where requirements are prioritised by the client at the beginning of each iteration, sprint backlogs where requirements are estimated by the development team, burn down charts that measures the work progress…). Ceremonies such as daily and retrospective meetings seem to enhance communication between team members (Overhage and Schlauderer, 2012; Sutherland et al., 2007; Sharp and Robinson, 2008; Chong, 2005; Melnik and Maurer, 2005; Robinson and Sharp, 2004). They help teams better control their projects and steer them towards their goals (Paasivaara et al., 2009; Vázquez-Bustelo and Avella, 2006). Iterative development adds agility to the development process by providing continuous feedback on the incremental product deliverable (Karlström and Runeson, 2005). It facilitates monitoring of the project's progress (Begel and Nagappan, 2007). It also enhances collaboration with customers and improves organisational learning by incorporating consumer feedback into future iterations (Middleton et al., 2005). Besides, agile collaboration tools such as storyboards and virtual whiteboards reinforce information exchanges between team members. These tools create an informative workspace and improve the common vision of the project (Khalil et al., 2013; Paasivaara et al., 2009).

The literature review highlights that practitioners have differing positions regarding the implementation of the 'right' project management methodology. Each methodology has its advantages and disadvantages. While more and more practitioners consider that agile methodologies can be used on large projects especially when agile principles are respected and adapted to the team's needs (Gallagher and Garratt, 2019; Jørgensen, 2018; Hobbs and Petit, 2017; Linders, 2016; Paasivaara et al., 2009), others believe that these methodologies merely fit small and simple software projects (Kettunen, 2009; Cohen et al., 2004). According to the latter group, plan-driven approaches are essential for complex environments and projects. Considering these divergent views, a group of researchers and practitioners questioned the combination of agile and traditional methodologies in order to better respond to the competitive environment and project needs (Rodov and Teixidó, 2016; Andersen, 2016; Sommer et al., 2015; Conforto and Amaral, 2010; Lozo and Jovanovic, 2009; Port and Bui, 2009; Wysocki, 2007).

2.2 Combining agile and plan-driven methodologies

Combining agile and traditional plan-driven methodologies seems to provide industrial companies with considerable performance benefits by enabling teams to better focus on business value and continuous resource allocation (Bianchi et al., 2020; Cegarra-Navarro et al., 2016; Sommer et al., 2015; Baird and Riggins, 2012; Barlow et al., 2010; Hass, 2007; Kahkonen, 2004). In addition, the integration of agile collaboration practices (daily ceremonies, collective code ownership) and visual management tools in plan- driven environments seems to improve team communication and knowledge sharing (Cegarra-Navarro et al., 2016; Sommer et al., 2015; Hass, 2007).

However, as highlighted by many researchers, it is essential to reach a balance between agile and plan- driven methodologies (Heeager, 2014; Mahnic and Zabkar, 2008; Boehm and Turner, 2003; Boehm, 2002). Several studies focus on determining the factors that affect the choice of project management methodology (Conforto et al., 2014; Balaji and Murugaiyan, 2012). According to many authors (Conforto et al., 2014; Spundak, 2014; Chin and Spowage, 2010; Cheema and Shahid, 2005; Highsmith, 2004; Boehm and Turner, 2003; Cockburn, 2000), several factors need to be assessed before determining the appropriate hybrid methodology including project size, project criticality, project innovation, and team and organisational characteristics. Even though a growing number of studies stress the need to mix agile and traditional methodologies, few have made recommendations on how to practically blend agile and plan-driven methodologies (Rodov and Teixidó, 2016; Sommer et al., 2015; Spundak, 2014; Lozo and Jovanovic, 2012; Hass, 2007).

In addition, traditional organisational structures and processes can constrain the integration of agile practices (Lohan et al., 2010). The recommendations highlighted in the literature evoke the need to ensure organisational transitions when integrating agile practices into plan-driven environments, building autonomous and self-managing development teams, involving top management in the implementation of hybrid methodologies, as well as improving communication between different stakeholders.

Despite the contributions of these studies to the project management domain, the implementation of hybrid models is still challenging and under-examined. Further research is still necessary. Teams lack structured processes and key success factors that help them integrate agile practices and principles into plan-driven environments. For instance, the required planning and documentation levels are difficult to obtain in hybrid projects. Besides, the integration of collaboration practices (scrum ceremonies, pair programming, and collective code ownership) in plan-driven environments is not always easy to achieve (Khalil and Fernandez, 2011, 2012). Research thus needs to conduct in-depth investigations into the barriers and success factors that should be taken into consideration when implementing a hybrid methodology. In this respect, this paper aims to fill this gap by proposing a model that combines agile and plan-driven methodologies. The success factors, risks and benefits associated with the suggested hybrid model are addressed in the Findings section.

3 Research methodology

3.1 Data collection

A qualitative research approach has been adopted in order to understand the way agile and plan-driven methodologies can be mixed from the point of view of our participants. Eighteen semi-structured interviews were conducted, each lasting an average of 2 h, with a panel composed of project management consultants with agile and traditional project management background and expertise.

While all of these consultants were members of the same consultancy group, each was attached to one of the group's three subsidiaries. Each subsidiary specialised in a specific field of activity:

- Subsidiary 1 specialised in software engineering and provided expertise in software and embedded systems;
- Subsidiary 2 specialised in quality and industrial performance and provided expertise in system and equipment quality assurance;
- Subsidiary 3 specialised in the field of complex project management and provided expertise in IT services and project organisation.

It is also important to note that each subsidiary has its own clients (EDF, SNCF, Airbus and so on) and consultants.

The projects in which our participants were involved thus differ in terms of human, technical and managerial characteristics. Table 1 presents the panel of participants, their positions, the subsidiary they are attached to, and the industrial sector they are currently working for.

Table 1 Participants detailed information

Consultant ref	*Subsidiary*	*Subsidiary Specialty*	*Position*	*Industry*	*Interview length (in minutes)*
C.1	Subsidiary 1	Software Engineering	Quality Software Manager	Energy	134
C.2			Project Manager	Aeronautics	121
C.3	Subsidiary 2	Quality and Industrial Performance	Engineering Technical director	Aeronautics	98
C.4			Program Manager	Aeronautics	128
C.5			Software Development Director	Transportation	110
C.6			Developer	Transportation	130
C.7			Project Manager Officer	Transportation	125
C.8			Developer	Transportation	132
C.9	Subsidiary 3	Complex Project Management	Project Manager	Aeronautics	123
C.10			Project Manager/ Scrum Master	Aeronautics	117
C.11			Project Director	Banking	125
C.12			Project Manager	Banking	118
C.13			Project Manager	Energy	123
C.14			Scrum Master/ Agile Coach	IT Provider	119
C.15			Agile Coach	IT Provider	115
C.16			Project Manager	Transportation	121
C.17			Project Director	Transportation	120
C.18			R&D Director	Services	118

Many questions were addressed during the interviews (the interview guide can be found in Appendix). The aim of these interviews was

1 to explore the organisational and managerial context in which each participant works
2 to examine the potential use of hybrid models for managing complex projects from the participants' perspectives
3 to understand their perceptions regarding the usefulness of combining agile and traditional project management methodologies as well as the best way(s) to do it.

Our participants were informed about the purpose of this study. Invitations to participate in the study were sent to all interviewees by the Research and Development director of subsidiary 3. In addition to the interviews, we had access to secondary data sources including organisational charts, Gantt charts, project reports, and newsletters that helped us learn more about each subsidiary. This data source was very helpful during data analysis. It helped us grasped the different contexts, practices, and impediments encountered in each subsidiary.

3.2 Data analysis

All the interviews were recorded with the consent of the participants, and then transcribed. The quotes presented in this research paper were translated from French to English. We adopted an interpretive approach when analysing the collected data. We began with multiple readings of the transcribed interviews to better understand the context and projects in which the respondents were involved. Through using the software NVivo 11 (version 11), an open-coding approach was adopted in order to identify key categories in each transcribed word, sentence and/or paragraph. Codes were generated, sorted, and refined throughout reading the transcripts. Subsequently a set of inductive categories and subcategories were defined iteratively and justified with verbatim quotes (Gibson and Brown, 2009; Miles and Huberman, 1994). Table 2 illustrates a sample of categories and their associated quotes.

4 Findings

Our findings demonstrate that organisations do not yet have a structured hybrid approach they can rely on to manage their projects. According to the interviewees, combining agile and plan-driven methodologies is not easy. On one side, the waterfall or 'V' lifecycle models constrain the integration of agile practices: "*If a team decides to integrate agile practices into their* "*V*" *cycle model, the team will likely be working according to the* '*V*' *cycle mode*l" (C.7). On the other side, viable software products can be delivered much more quickly in agile environments than traditional environments. Mixing agile and traditional projects can thus constrain agile teams who need to integrate their work frequently and deliver working products rapidly: "*Through iterative development and continuous integration, the agile team was able to deliver a viable product that had reached a higher maturity level compared to the one delivered by the traditional team*" (C.9). Although combining both methodologies seems challenging, the respondents stressed the need to have a structured hybrid methodology that helps them manage complex projects and balance the agile and traditional environment. The aim of a mixed methodology is to balance the limitations of both traditional and agile approaches. That said, such a mixed methodology still needs to be tested and adapted to the organisation's and team's needs.

Table 2 Sample of data analysis

Categories	*Quotes*
Perceived benefits of agile practices	"*This* (*iterative development and integration*) *will help us identify potential defects as early as possible*" (C.16)
Combining traditional and agile methodologies	"*What I have noticed during my work with different clients is that agile and traditional methodologies can be combine*d" (C.13)
Hybridisation challenges – synchronisation challenge between traditional and agile teams	"*Many barriers can arise. First of all, the synchronisation of agile and traditional development teams*". *Let's take the example of this eighteen months' project where agile and traditional teams were both involved in the product development and delivery. Through iterative development and continuous integration, the agile team was able to deliver a viable product that had reached a higher maturity level compared to the one delivered by the traditional team*" (C.12)
A proposed hybrid model (3): creating mini traditional cycles	"*The iterations are apparent to mini waterfall cycles. However, the duration of each cycle is eight to twelve weeks… I know it is more than what agile or scrum recommends…*" (C.3)
Factor of success associated with the mini traditional cycle model	"*We have to timebox each mini V or waterfall cycle*" (C.2)

The sections below portray the main characteristics of projects where each of the suggested hybrid models can be used. It also details the associated benefits, challenges and success factors for each model as suggested by the participants.

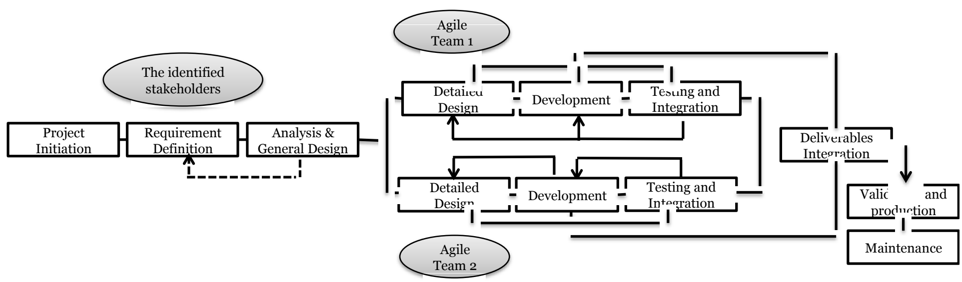
4.1 Hybrid model 1

Hybrid model n°1: Decomposing the project into both sequential and iterative phases that integrate agile practices (Figure 1).

Even though this model converges with existing iterative and incremental frameworks and hybrid models found in the literature (Rahmamian, 2014; Lozo and Jovanovic, 2009), it offers a complete vision of how to mix traditional with agile methodologies. The respondents who proposed and discussed this model believe that the project initiation, requirement definition and architectural design phases should be linear and conducted in a traditional way. The project scope, budget, schedule as well as major requirements are therefore defined up-front with the participation of different stakeholders.

In addition, the main structural components in the system and the relationships between them are also identified. That said, architectural decomposition can be used by engineering teams as a 'mean' to re- discuss the requirements of the system. It is thus possible to step back to the requirement definition phase if engineering challenges are identified in the general design phase. Thereafter the detailed design, development and testing and integration phases are iterative. These phases are managed according to scrum principles along with collaboration practices and tools. Depending on the size of the project, the agile part can be assigned to one or several agile development teams. After the integration of the deliverables, the verification and production phases are conducted in a traditional way. The project manager is responsible of ensuring communication between different stakeholders. They are also in charge of ensuring that timeboxed iterations and meetings are respected. A product owner could assist the development team in order to prioritise requirements during iteration cycles and to validate the deliverables.

Figure 1 Hybrid model 1



4.1.1 *For which type of projects is this model most suited?*

This model has been used in the transportation industry for complex projects with the following characteristics: long durations (between two and three years), high levels of criticality (embedded technologies, normative constraints), high levels of innovation, moderate need for traceability, autonomous development teams not necessarily working for the same organisation in charge of project initiation and definition, and a client representative located off-site who is ready to collaborate when needed.

4.1.2 *The associated benefits*

On the one hand, this process enables teams to extensively plan their projects, define the major requirements up-front, and fix the project's overall architecture. This can be useful when "*the project is characterised by a high level of criticality and the client requires detailed plans and specifications*" (C.15). It also provides reassurance for project managers who often need "*detailed plans in order to control and monitor their projects*" (C.8). In addition, stabilising the overall architecture is helpful for reducing the costs associated with refactoring. It also helps project teams establish a global vision of what needs to be done.

On the other hand, the iterative design, development and testing phases will help the agile team focus on prioritised functionalities, improve code quality and integrate changes at a lower cost "*this will really help us getting constant feedback and make rapid changes*" (C.10). Errors will be detected sooner in the process and teams can better deal with unpredictable events due to the innovative aspect of the project. Besides, the scope of deliverables is minimised which decreases the associated risks. The hybrid model also helps reduce silos between the design, development and testing teams "*today each team works independently [...] it would be great to boost the collaboration between these teams*" (C.7).

4.1.3 The challenges

In this model, documentation costs are not minimised because the up-front phases require detailed plans and specifications. Accordingly, integrating changes would increase project costs and lead to rework. Also, misunderstandings can result from the lack of communication between agile teams and other groups (teams involved in the up-front phases as well as in the validation and production phases). Besides, coordinating several agile teams can be challenging.

4.1.4 The success factors

Several success factors emerge for the first hybrid model that consists of decomposing the project into both sequential and iterative phases integrating agile practices. First of all, it is critical for the design, development and production teams to be involved in the requirement definition phase. In addition, developers and testers should be part of the same agile team. Similarly, the collaboration between the development and production teams should be encouraged by the organisation. It is also very essential to have stable agile development teams while ensuring coordination between different agile teams through implementing collaboration tools, encouraging regular meetings, and so on. While the product owner role is important for prioritising requirements in the agile development cycles, the project manager is essential for being aware of agile thinking and iterative development process. Moreover, feedback should be encouraged at the end of each iterative cycle; "*we lack feedback on the work we do. I think this should be encouraged*" (C.3). Although the major requirements are defined up-front, it is important to have the client's feedback regarding each integrated deliverable.

4.2 Hybrid model 2

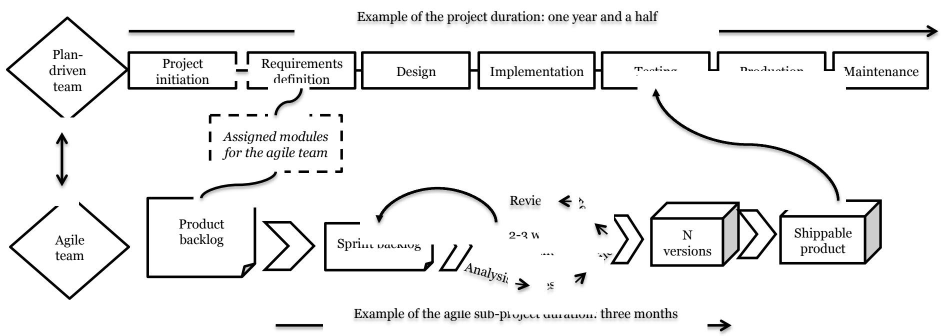
Hybrid model n°2: Decomposing the project into two independent sub-projects: an agile sub-project and a traditional sub-project (Figure 2).

This hybrid model was proposed by five of our respondents who work in two different subsidiaries. They stated that this model seems to fit large organisations that have both agile and traditional teams who work independently but can occasionally be involved in the same project. The project is therefore split between the two teams. While the traditional team is in charge of developing their part of the product and delivering the intended final product to the customer, the agile development team works on their part of the project according to agile principles and practices. Their deliverables are integrated into the product developed by the traditional team. The agile team thus intervenes promptly in the project. The agile development process starts once the modules/ components are defined and assigned to the agile development team(s). The duration of both traditional and agile projects can vary.

4.2.1 For which type of projects is this model most suited?

Our interviewees believed that this model could be helpful for matrix organisational structures that involve traditional teams that value plan-driven methodologies and hierarchical functioning on the one side, and agile development teams emphasising agile principles (autonomous and polyvalent team members, iterative work, adaptation to changes and so on) on the other. The participants stated that the proposed hybrid model "*has been and/or would be applied in large IT projects characterised by cross-functional team members who are attached to different functional managers*" (C.6) and also "*work, for a given period of time, under the authority of a traditional project manager who values stage-gate methodologies*" (C.9). However, this model is not limited to the telecommunication sector. It can also be used in other industries, as well as other organisational structures where agile and traditional teams are brought to work together.

Figure 2 Hybrid model 2



4.2.2 The associated benefits

This model can balance the documentation and the standardisation needs of agile and traditional teams. On one hand, traditional teams can rely on a formal process, plan it and document it. The schedule and budget along with the lists of features and resources are fixed up-front "*we need to do all the planning up-front, it's inevitable*" (C.1). In addition, components with high-critical levels are specified and documented according to defined standards and norms "*critical products need to be documented*" (C.16). On the other side, the agile team will reduce the documentation and increase code quality through iterative development, unit-testing, continuous integration, collective code-ownership, pair programming and so forth. Agile practices can thus be helpful for "*innovative modules or components of the overall project*" (C.4). Besides, the agile team will boost the waterfall project by delivering, iteratively, working increments.

4.2.3 The challenges

This process necessitates coordination between both teams, which can be challenging. While the traditional team relies on reports and weekly meetings, the agile team emphasises daily meetings, mutual adjustment and visual tools. Besides, synchronisation between both teams "*is not easy to achieve*" (C.11). Change integration in the agile process will generate more work for the traditional team (more specifications, tests and so on), which can be challenging on the resource leveling "*integrating changes generates more work for our team*" (C.10). For instance, the traditional qualification team will need to anticipate the qualification phase by writing tests along with the agile sprints.

However, according to our participants this happens infrequently, leading to delays and misunderstandings between teams. Conflicts of interest and competition among teams may also emerge in such hybrid projects. In addition, managing agile teams can be challenging for project managers who tend to perceive auto- organised teams as a threat to their position. Finally, agile teams can feel frustrated as they intervene punctually in the project, and rarely have the occasion to monitor the final product "*it's somewhat frustrating not be able to be part of the final product*" (C.3).

4.2.4 The success factors

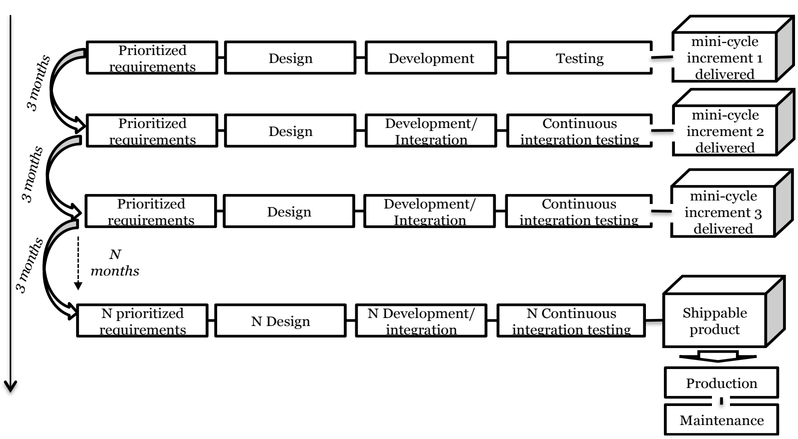
For the second hybrid model, decomposing the project into two independent sub-projects – an agile sub- project and a traditional sub-project, the following success factors emerged. First, the plan driven team and the agile team should agree on common conceptual architecture models, programming languages, coding norms, and so on. Once they agree, a common programming tool should be deployed and shared by both teams. Aside from sharing common tools, both teams should also share a database to instore transparency where all their data are stored. In addition, while meetings between both teams should be organised where they can share project progress and improve their common vision, change integration in the agile development process should be communicated to the traditional team. Finally, the project manager should always be aware of agile principles and mindsets.

4.3 Hybrid model 3

Hybrid model n°3: Decomposing the waterfall or V-cycle model into small mini waterfall or V-cycles (three-months duration) (Figure 3).

The aim of this model is to integrate iterative thinking in the 'V' or waterfall lifecycle model and so decrease the project lifecycle duration. As in traditional lifecycles, each mini-cycle consists of different linear phases such as requirement definition, design, development, testing and integration. The duration of each lifecycle is between two and three months. Project team members are attached to different functional services.

Figure 3 Hybrid model 3



4.3.1 For which type of projects is this model most suited?

Our interviewees believe that this model can be useful for large and complex projects, for example in the aeronautics and space industries. These projects are characterised by very high levels of criticality, intensive security regulations, and extensive documentation and documentation standards.

4.3.2 The associated benefits

This model can be used as an alternative way to manage complex projects that require a structured linear approach. It fits projects with high criticality levels and documentation standards. Projects are divided into smaller projects, making them easier to manage. Even though documents are still delivered at the end of each linear phase and mini-cycle, the risk of producing obsolete documents is mitigated because of the mini-cycle's limited scope "*by having small cycles, we will document less*" (C.18). The iterative mini-cycles enhance feedback regarding the delivered working increment.

The client's representative has an important role to play at the beginning and end of each mini-cycle. They prioritise the requirements that need to be implemented in each mini-cycle and give feedback regarding each delivered version of the product. Even though each cycle lasts three months, instead of up to four weeks as recommended in agile methodologies, these iterative cycles still improve the quality of the delivered versions; "*we can better monitor our deliverables*" (C.2). In addition, this iterative way of working decreases the risks associated with up-front planning and requirement definition. Requirements can be reviewed and reprioritised at the end of each mini cycle, which facilitates change integration and decreases the associated costs. According to our interviewees, this model can improve relations with the client as the latter is supposed to intervene more often in the process.

4.3.3 The challenges

Unlike agile projects with three-month-long sprints where team members are polyvalent, this model does not eliminate silos between functional teams. In addition, extensive documentation generates waste even through the risk of producing obsolete documents is minimised. Furthermore, it is difficult to involve the client throughout the mini cycles as mentioned by this consultant; "*we need to shift our client's mindset. I am not sure our client will agree to participate in all these meetings*" (C.7). The client relationship is thus dissimilar to the one in an agile project. Besides, stabilising the mini cycle team is challenging. The project manager needs to have enough authority to negotiate their functional resources and stabilise their team through iterations. According to our respondents, functional team members can be involved in different projects in parallel, which can complicate their intervention in each iterative cycle. Unstable teams can thus affect knowledge sharing as well as the project vision, leading to relearning, waiting times and delays.

4.3.4 The success factors

Numerous success factors emerged for the third hybrid model – decomposing the waterfall or V-cycle model into small mini waterfall or V-cycles. First of all, in order to negotiate their resources and stabilise their teams, project managers need to have authority with their teams. It is also extremely important for the top management to support the authority of project managers. In addition, "*project managers should be aware of the agile mindset and the ways of working*" (C.14). Moreover, knowledge capitalisation is essential especially when functional team members are unstable. While lifecycles should stay short and not exceed three months, retrospective meetings should always be encouraged between the team and the client's representative. Finally, multidisciplinary teams are motivated to be divided into 'siloed' functional teams. The following Table 3 summarises the three hybrid models that emerged from our analysis, illustrating their own objectives, the most suitable types of projects, their benefits, their challenges and their success factor.

Table 3 Hybrid models summary

Hybrid Model	*Objective*	*Type ofprojects*	*Benefits*	*Challenges*	*Success factors*
Hybrid Model 1	Decomposing the projectinto both sequential and iterative phases that integrate	• Long durations • High levels of criticality • High levels of innovation	• Extensively plan projects • Defining major requirements up-front	• Documentation costs not minimised • Increased project costs • Lack of communication	• Developers and testers belonging to same agile team • Collaboration between development and production teams
	Agile practices	• Moderate need for traceability	• Fixed project's overall architecture		• Feedback • Agile thinking and iterative development process
Hybrid Model 2	Decomposing the projectinto two independent sub-projects:an agile sub-project and atraditional sub-project	• Matrix organisational structures • Large IT projects	• Planning and documenting formal process • Fixed schedule and budget up-front • Components with high-critical levels documented according to defined standards and norms	• Coordination between both teams • Synchronisation • Change integration	• Common conceptual architecture models, programming language and tools, coding norms • Meetings between both teams • Agile principles and mindsets • Shared database
Hybrid Model 3	Decomposing the waterfall or V-cycle model into small mini waterfall or V-cycles (three-months duration)	• Large and complex projects (aeronautics, space projects)	• Projects with high criticality levels and documentation standards • Feedback • Client's representative • Iterative way of working	• Does not eliminate silos between functional teams • Waste generated • Difficulty to involve the client throughout the mini cycles • Hard to stabilise the mini cycle team	• Project manager authority • Knowledge capitalisation • Short lifecycles • Multidisciplinary teams • Retrospective meetings

5 Discussion and conclusion

Based on our interview analyses, three hybrid models emerged. The first hybrid model consists of decomposing the project into sequential and iterative phases that integrate agile practices. The participants stress that this model offers a complete vision to mix traditional and agile methodologies. As suggested by Conforto et al. (2014), Spundak (2014) and Chin and Spowage (2010), several factors need to be assessed before choosing the appropriate hybrid methodology. In this case, this hybrid model can be a good fit for projects with long durations, high levels of criticality and, high levels of innovation (Heeager, 2014; Mahnic and Zabkar, 2008; Boehm and Turner, 2003). The second emerging model consists of decomposing the project into independent sub-projects; an agile sub-project as well as a traditional one. This model is best fit for large organisations having agile and traditional teams that work independently and occasionally on the same project. However, this model does not seem to tackle the problems encountered by teams using traditional methodologies. A third model emerged from our analyses. It consists on decomposing the waterfall (or V-cycle) model into smaller mini waterfalls (or V-cycles). The objective of this model is to combine the iterative thinking of the V-model with a decreased lifecycle duration. However, according to the participants, the production seems to happen at the end instead of having multiple deliveries as recommended in agile environments (Savor et al., 2016). This reinforces the need for testing these suggested models in different environments.

Furthermore, according to the interviewees, many contextual factors should be assessed and taken into consideration before implementing the right hybrid methodology. Among these factors, we identified: project size, project criticality, project manager personality, team composition, level of customer trust, customer availability, ease of requirement definition, organisational structure and culture, as well as project technical component interdependencies. These factors seem to have an impact on the choice of hybrid model. They affect documentation needs, planning levels, collaboration levels between team members and with the client, project lifecycle duration, knowledge capitalisation systems and so on.

As analysis of the data has shown, combining agile and plan-driven methodologies is perceived as helpful for managing complex projects. Agile and traditional approaches can be combined in different ways. This research paper described and analysed three hybrid models that were proposed and discussed by our participants. Each model has its advantages, challenges and success factors. However, the choice of appropriate hybrid model depends on the project type and various contextual factors (project structure, stakeholders involved, teams' organisation, project manager authority and so on). Given the different contexts, our analyses help differentiate each model for the most suitable contexts.

5.1 Theoretical implications

Few research papers focused on how to combine agile and traditional methodologies. This study has made two major implications to the project management and agility literature. First, this paper provides three different theoretical frameworks for combining agile and traditional project management approaches. These frameworks offer new perspectives on how to plan, execute and control projects based on the type of projects and context settings. They contribute to the 'non-uniform nature' of project management research. Our qualitative research methodology helped us understand the context and factors that should be taken into consideration when projects are executed. Hence, projects are viewed as complex social systems where different dimensions (organisation/ structure/culture/uncertainty/complexity/duration...) need to be considered in order to identify the most appropriate mixed project management approach.

Second, this piece offers new insights to the agile literature. Our analyses present three hybrid frameworks that can enable organisations to adopt agile methodologies. These frameworks contribute to the agile literature through highlighting that short and iterative development lifecycles, ongoing collaboration and adaptation to changing requirements remain at the core of agile practices when combined with plan-driven methodologies.

5.2 Practical contributions

Our research paper provides contributions to practitioners. Our analyses portray three practical hybrid approaches which project teams can rely on to manage their complex projects. Given that (to our knowledge) teams within organisations lack a well-defined hybrid approach that will help them combine agile and plan-driven methodologies, our research paper allows teams to base their work on one of the three proposed models. Several hybrid models were developed recently and used within large organisations. These models emphasised collaboration culture and network (Spotify model) (Kniberg and Ivarsson, 2012) and focused on scaling agile methods to large projects including combining lean, agile and DevOps practices (i.e., SAFe) (Marinho et al., 2021) as well as scaling scrum to enhance cross-team coordination (i.e., SoS) (Paasivaara et al., 2012). Learning about these models would have provided additional input to the three suggested hybrid models, leading to more agility to the project management practices within each subsidiary. However, the participants were not aware of these large-scale agile frameworks which have obviously influenced their mental models.

5.3 Limitations and future research

This paper presents four main limitations: the suggested hybrid models as well as their success factors require more investigations. Although the proposed hybrid models are represented as three alternative ways to cope with complex projects, they still need to be used on more projects in order to investigate the associated challenges and success factors. Additionally, our participants did not have strong background and knowledge with agile methods and large-scale agile frameworks models such as Spotify Model, LeSS, SAFe, etc. They had more experience in traditional methodologies which might have influenced the emergence of the suggested hybrid models.

While the interviews took place in a French context, additional observations and interviews in multiple project settings would also help to test the findings and generalise them.

Furthermore, another limitation is the number of our panel of participants. For further investigations, it would be interesting to apply the three hybrid models on a bigger panel.

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Appendix

1 How do you characterise the projects in which you have been involved as a consultant (in terms of size, duration, complexity levels, main constraints, etc.)?

2 Can you please tell us about the project management methodologies that have been used in these projects?

3 What were the main challenges encountered in these projects?

4 Can you please explain the main reasons for the encountered challenges?

5 According to you, what factors affect the choice of project management methodology?

6 In what type(s) of projects are agile methodologies the best fit? Would you please tell us why?

7 In what type(s) of projects are plan-driven methodologies the best fit? Would you please tell us why?

8 What challenges do agile teams face?

9 What challenges do traditional teams face?

10 Have you ever worked on projects where agile and traditional methodologies are combined?

   i If yes, would you give us one or several examples of mixing these two methodologies?

   ii If no, what are your thoughts regarding mixing agile and plan-driven methodologies? Theoretically how would a hybrid approach be possible?

11 In what situations do agile practices support plan-driven teams? (And vice-versa)

12 Which success/failure factors should be taken into consideration when combining these two methodologies?